

```

%% PHYS 5394 Lab: Signals and Periodograms
% This live script generates all eight lab signals and displays each signal
% beside the magnitude of its positive-frequency FFT.
%
% The eight crcbgen... function files must be in the Current Folder.

clear;
close all;
clc;

% Common matched-filtering SNR
snr = 10;

% Common observation duration
signalDuration = 1.0;

%% 1. Quadratic chirp
% The coefficients produce a visible nonlinear frequency increase from
% 10 Hz to 25 Hz during the one-second observation.

a1 = 10;
a2 = 3;
a3 = 3;

% Maximum instantaneous frequency:
%  $f(t) = a_1 + 2*a_2*t + 3*a_3*t^2$ 
maxFreq = a1 ...
    + 2*a2*signalDuration ...
    + 3*a3*signalDuration^2;

% Sample at ten times the estimated maximum frequency
samplFreq = 10*maxFreq;

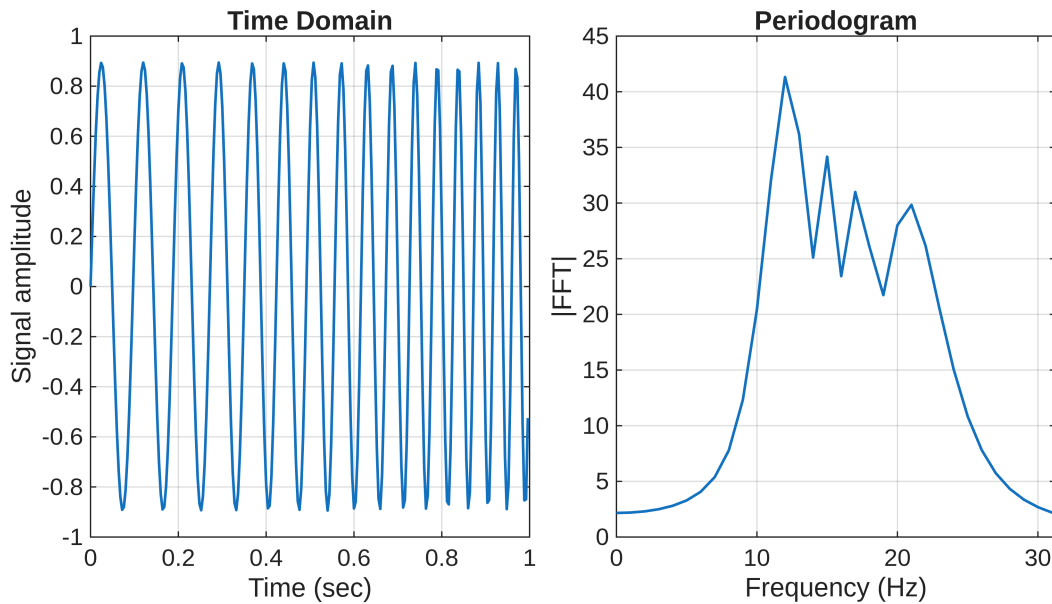
timeVec = makeTimeVector(samplFreq,signalDuration);

sigVec = crcbgenqcsig(timeVec,snr,[a1,a2,a3]);

showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'Quadratic Chirp',1.25*maxFreq);

```

Quadratic Chirp



```
fprintf(['Quadratic chirp: its broad 10-25 Hz spectrum represents a ' ...
        'nonlinear increase of instantaneous frequency.\n\n']);
```

Quadratic chirp: its broad 10-25 Hz spectrum represents a nonlinear increase of instantaneous frequency.

```
%% 2. Sinusoidal signal
% A 10 Hz tone over one second lies exactly on a 1 Hz DFT bin, providing
% a simple reference spectrum with minimal spectral leakage.
```

```
f0 = 10;
phi0 = 0;

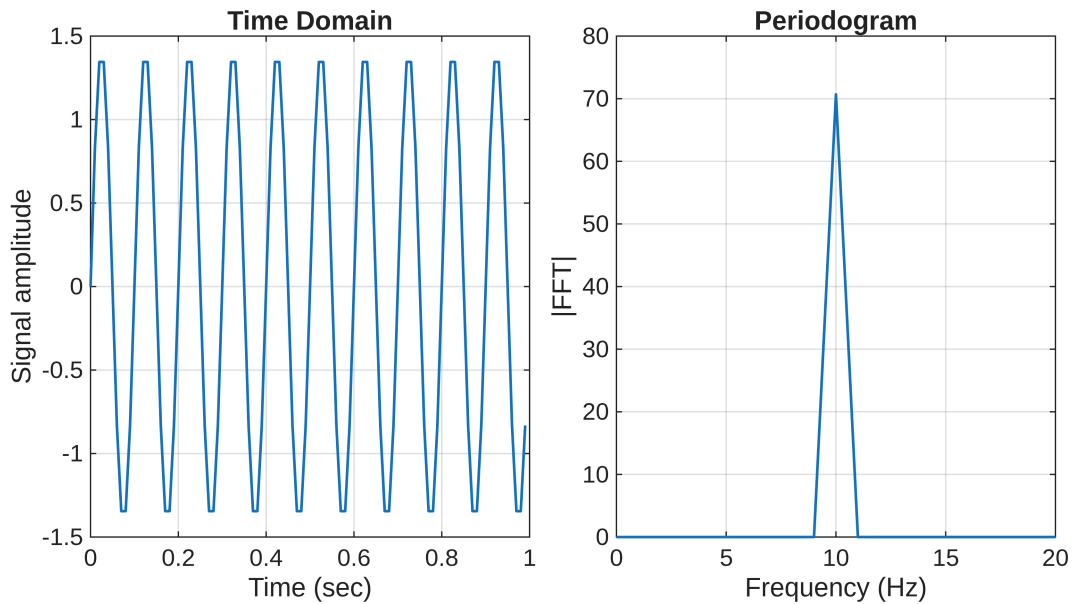
maxFreq = f0;
samplFreq = 10*maxFreq;

timeVec = makeTimeVector(samplFreq,signalDuration);

sigVec = crcbgensinsig(timeVec,snr,[f0,phi0]);

showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'Sinusoidal Signal',2*maxFreq);
```

Sinusoidal Signal



```
fprintf(['Sinusoid: the narrow peak at 10 Hz represents a signal with one '
...
'constant frequency.\n\n']);
```

Sinusoid: the narrow peak at 10 Hz represents a signal with one constant frequency.

```
%% 3. Linear chirp
% These parameters produce a uniform frequency sweep from 10 Hz to 30 Hz
% during the one-second observation.
```

```
f0 = 10;
f1 = 20;
phi0 = 0;

initialFreq = f0;
finalFreq = f0 + f1*signalDuration;

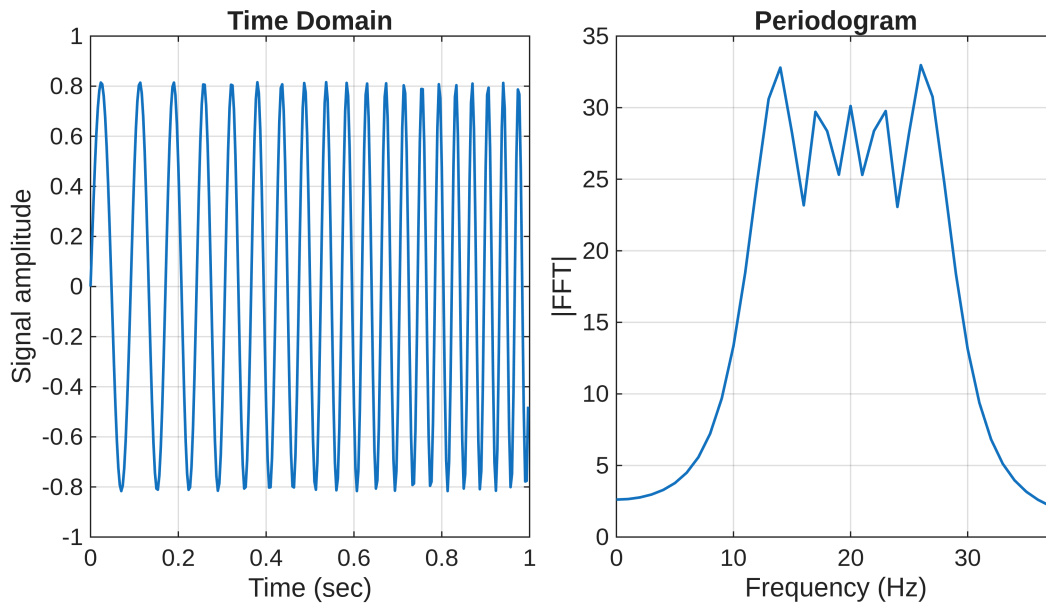
maxFreq = max(abs([initialFreq,finalFreq]));
samplFreq = 10*maxFreq;

timeVec = makeTimeVector(samplFreq,signalDuration);

sigVec = crcbgenlcsig(timeVec,snr,[f0,f1,phi0]);

showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
'Linear Chirp',1.25*maxFreq);
```

Linear Chirp



```
fprintf(['Linear chirp: energy distributed from about 10-30 Hz represents '
... 'a constant rate of frequency increase.\n\n']);
```

Linear chirp: energy distributed from about 10-30 Hz represents a constant rate of frequency increase.

```
%% 4. Sine-Gaussian signal
% The pulse is centered at 0.5 s. A width of sigma = 0.1 s localizes it
% in time while preserving a visible band around the 30 Hz carrier.
```

```
t0 = 0.5;
sigma = 0.1;
f0 = 30;
phi0 = 0;
```

```
% The frequency-domain standard deviation of the Gaussian is:
% sigma_f = 1/(2*pi*sigma)
freqStd = 1/(2*pi*sigma);
```

```
% Include three frequency-domain standard deviations
maxFreq = f0 + 3*freqStd;
```

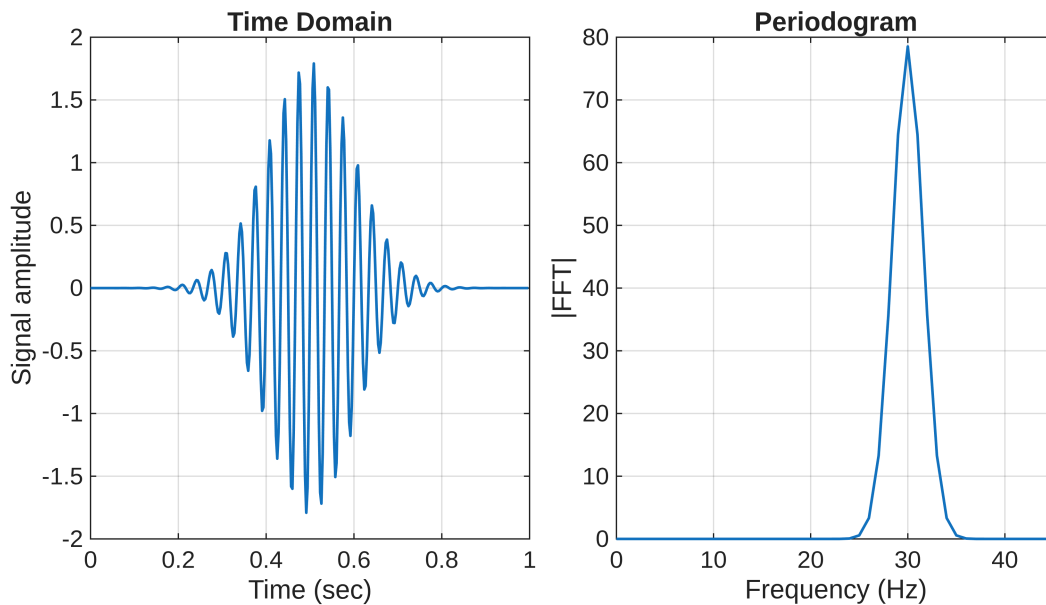
```
samplFreq = ceil(10*maxFreq);
```

```
timeVec = makeTimeVector(samplFreq,signalDuration);
```

```
sigVec = crcbgensgsig(timeVec,snr,[t0,sigma,f0,phi0]);
```

```
showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'Sine-Gaussian Signal',1.3*maxFreq);
```

Sine-Gaussian Signal



```
fprintf(['Sine-Gaussian: the Gaussian-shaped band near 30 Hz reflects the '
    ...
    'time-frequency tradeoff of a localized burst.\n\n']);
```

Sine-Gaussian: the Gaussian-shaped band near 30 Hz reflects the time-frequency tradeoff of a localized burst

```
%% 5. Frequency-modulated sinusoid
% b = 4 produces a visible frequency deviation, while f1 = 5 Hz gives
% clearly separated sidebands around the 40 Hz carrier.
```

```
b = 4;
f0 = 40;
f1 = 5;
```

```
% Peak FM frequency deviation
deltaF = abs(b)*f1;
```

```
% Carson-rule estimate of the highest significant frequency
maxFreq = f0 + deltaF + f1;
```

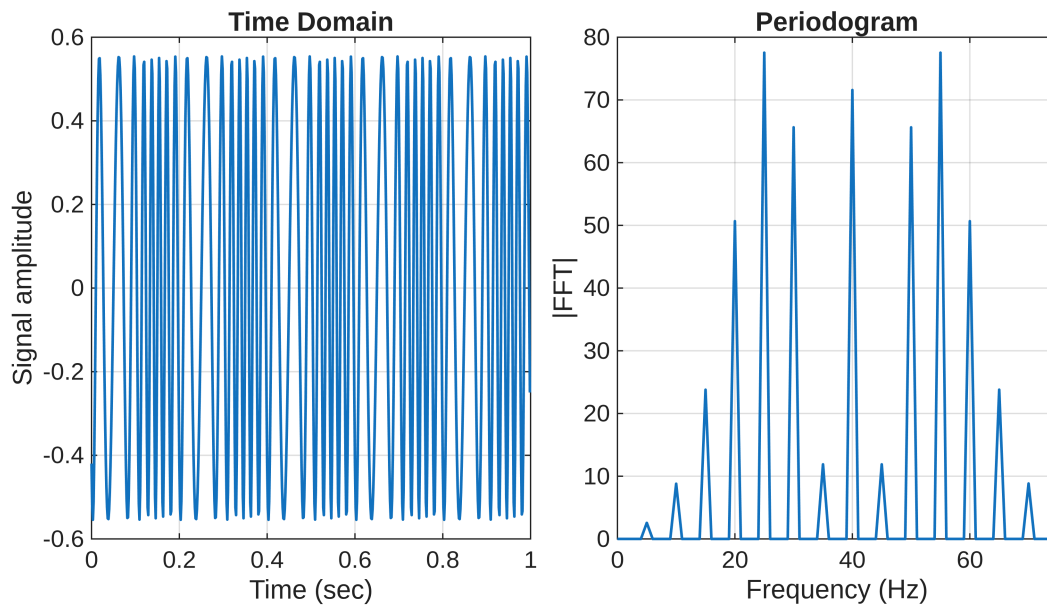
```
samplFreq = 10*maxFreq;
```

```
timeVec = makeTimeVector(samplFreq,signalDuration);
```

```
sigVec = crcbgenfmsig(timeVec,snr,[b,f0,f1]);
```

```
showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'FM Sinusoid',1.15*maxFreq);
```

FM Sinusoid



```
fprintf(['FM sinusoid: sidebands spaced by 5 Hz show how periodic frequency
' ...
'modulation redistributes carrier energy.\n\n']);
```

FM sinusoid: sidebands spaced by 5 Hz show how periodic frequency modulation redistributes carrier energy.

```
%% 6. Amplitude-modulated sinusoid
% A 40 Hz carrier and 5 Hz modulation produce easily identified lower
% and upper sidebands at 35 Hz and 45 Hz.
```

```
f0 = 40;
f1 = 5;
phi0 = 0;
```

```
% Highest AM sideband
maxFreq = f0 + f1;
```

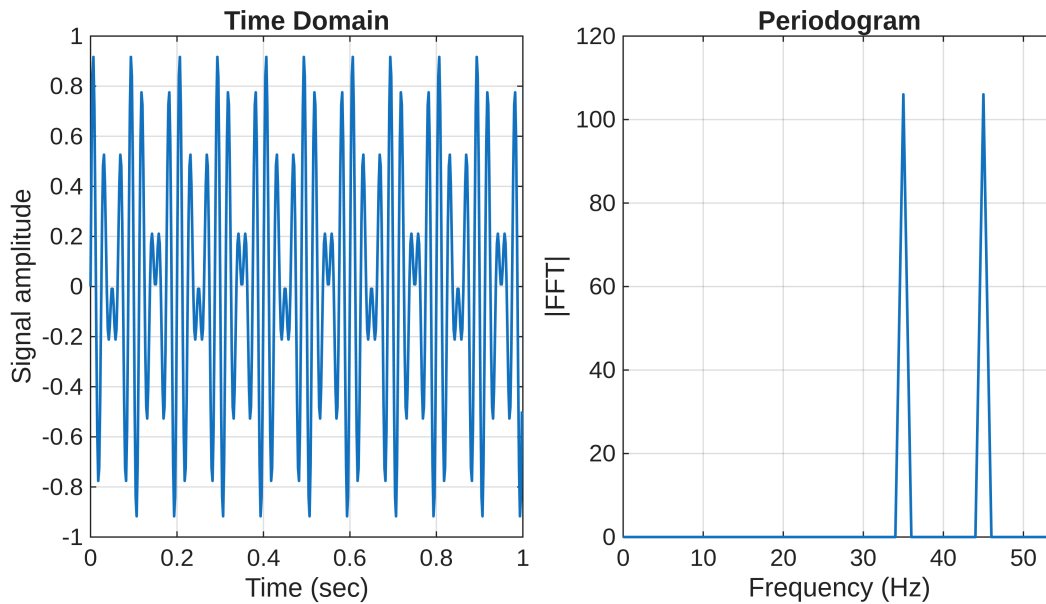
```
samplFreq = 10*maxFreq;
```

```
timeVec = makeTimeVector(samplFreq,signalDuration);
```

```
sigVec = crcbgenamsig(timeVec,snr,[f0,f1,phi0]);
```

```
showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'AM Sinusoid',1.2*maxFreq);
```

AM Sinusoid



```
fprintf(['AM sinusoid: peaks at 35 Hz and 45 Hz are the lower and upper ' ...
        'sidebands of suppressed-carrier amplitude modulation.\n\n']);
```

AM sinusoid: peaks at 35 Hz and 45 Hz are the lower and upper sidebands of suppressed-carrier amplitude modulation.

```
%% 7. AM-FM sinusoid
% The same values used in the separate AM and FM examples make their
% combined effects easy to compare.

b = 4;
f0 = 40;
f1 = 5;

deltaF = abs(b)*f1;

% FM bandwidth plus an additional frequency shift from the AM term
maxFreq = f0 + deltaF + 2*f1;

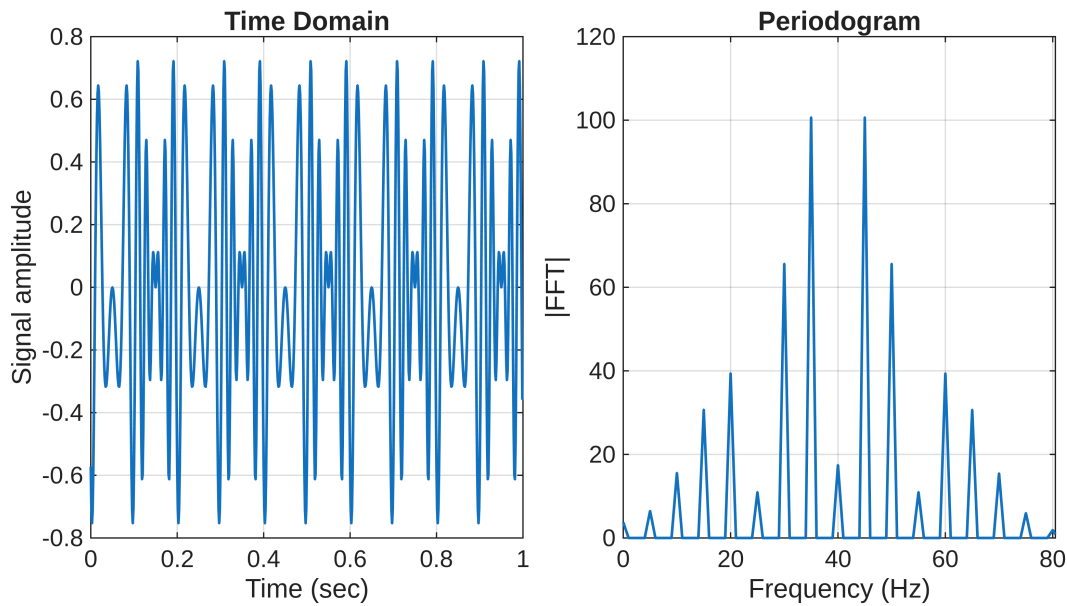
samplFreq = 10*maxFreq;

timeVec = makeTimeVector(samplFreq,signalDuration);

sigVec = crcbgenamfmsig(timeVec,snr,[b,f0,f1]);

showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
    'AM-FM Sinusoid',1.15*maxFreq);
```

AM-FM Sinusoid



```
fprintf(['AM-FM sinusoid: 5 Hz-spaced components with altered amplitudes '
...
'show the combined effects of amplitude and frequency modulation.\n\n']);
```

AM-FM sinusoid: 5 Hz-spaced components with altered amplitudes show the combined effects of amplitude and

```
%% 8. Linear transient chirp
% The chirp is active only from 0.2 s to 0.7 s and sweeps from 10 Hz to
% 30 Hz, making its finite duration and frequency evolution visible.
```

```
ta = 0.2;
f0 = 10;
f1 = 20;
phi0 = 0;
L = 0.5;

initialFreq = f0;
finalFreq = f0 + 2*f1*L;

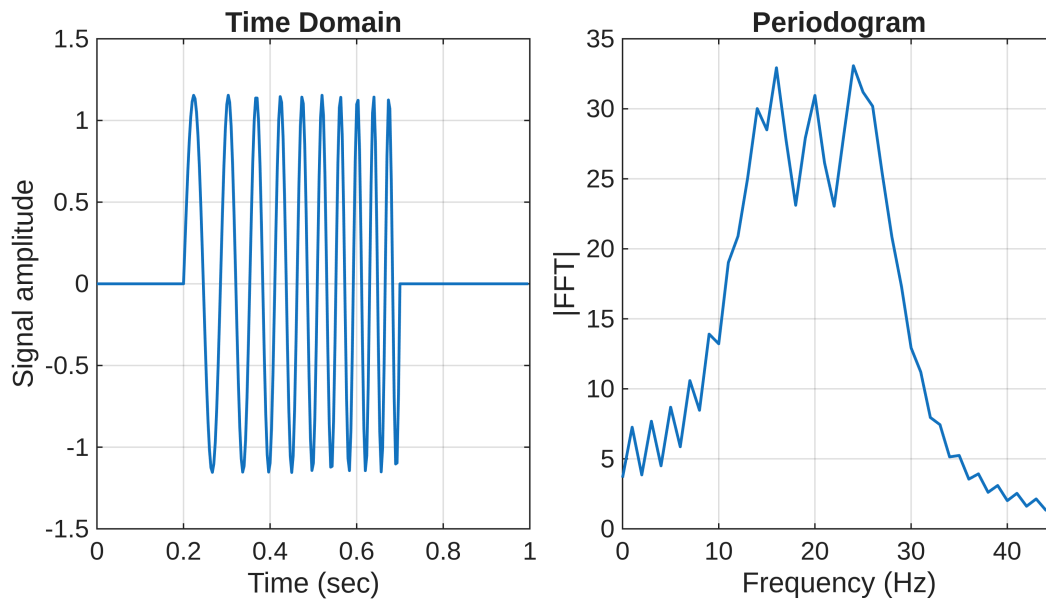
maxFreq = max(abs([initialFreq,finalFreq]));
samplFreq = 10*maxFreq;

timeVec = makeTimeVector(samplFreq,signalDuration);

sigVec = crcbgenltcsig(timeVec,snr,[ta,f0,f1,phi0,L]);

showSignalPeriodogram(timeVec,sigVec,samplFreq, ...
'Linear Transient Chirp',1.5*maxFreq);
```

Linear Transient Chirp



```
fprintf(['Transient chirp: broad 10-30 Hz energy plus leakage represents a '
...
'frequency sweep that is abruptly limited in time.\n\n']);
```

Transient chirp: broad 10-30 Hz energy plus leakage represents a frequency sweep that is abruptly limited

```
%% Local functions
```

```
function timeVec = makeTimeVector(samplFreq,signalDuration)
% MAKETIMEVECTOR generates samples over the interval 0 <= t < T.
%
% The time samples follow:
%
%     t_n = n/Fs,   n = 0,1,...,N-1.

nSamples = round(samplFreq*signalDuration);

timeVec = (0:(nSamples-1))/samplFreq;

end

function showSignalPeriodogram( ...
    timeVec,sigVec,samplFreq,signalName,maxPlotFreq)
% SHOWSIGNALPERIODOGRAM plots a signal and its positive-frequency FFT.

% Number of samples
nSamples = length(timeVec);
```

```

% Index of the highest nonnegative DFT frequency
kNyq = floor(nSamples/2)+1;

% Exact DFT frequency resolution
freqRes = samplFreq/nSamples;

% Nonnegative-frequency vector
posFreq = (0:(kNyq-1))*freqRes;

% Compute the FFT
fftSig = fft(sigVec);

% Periodogram: magnitude of the FFT
periodogram = abs(fftSig(1:kNyq));

% Create one time-domain and one frequency-domain panel
figure;

tiledlayout(1,2, ...
    'TileSpacing','compact', ...
    'Padding','compact');

% Time-domain signal
nexttile;

plot(timeVec,sigVec,'LineWidth',1.0);

xlabel('Time (sec)');
ylabel('Signal amplitude');
title('Time Domain');
grid on;

% Periodogram
nexttile;

plot(posFreq,periodogram,'LineWidth',1.0);

xlim([0,maxPlotFreq]);

xlabel('Frequency (Hz)');
ylabel('|FFT|');
title('Periodogram');
grid on;

% Figure title
sgtitle(signalName);

end

```